Hriday Adani

803-357-9673 | hriday1136@gmail.com | linkedin.com/in/hridayadani | github.com/hriday1136

Education

Rutgers University - New Brunswick — New Brunswick, NJ
Bachelor of Science; Computer Science and Economics — *Expected Graduation - May 2027*
Relevant Coursework : Data Structures, Artificial Intelligence, Algorithms

Projects

StorageKV | *Rust (zero runtime dependencies)*

- Engineered an embedded RocksDB/LevelDB-inspired key-value storage engine from scratch on an LSM-tree architecture with a WAL, immutable sorted SSTables with sparse indexing, and size-tiered compaction
- Implemented crash recovery via manifest checkpoint plus WAL replay, enforcing fsync ordering and atomic file installation so any acknowledged write survives crashes in the write path, similar to PostgreSQL's architecture
- Verified correctness through property testing against a reference model across **300+** randomized operation sequences with interleaved crash recovery, plus crash window tests for every durable operation
- Measured **sub-12μs p99** reads and quantified the cost of durability at approximately **1000x** over an in-memory baseline by isolating fsync overhead in same-machine benchmarks

ClauseWatch | *Next.js, FastAPI, PostgreSQL, OpenAI*

- Extracted structured fields from PDF and DOCX contracts at **97.9%** field-level accuracy (100% on effective and renewal dates) across a validated 30-document test set, using confidence-based OpenAI structured outputs
- Engineered 5 independently-secured HMAC token systems including webhooks, downloads, password reset, email verification, and cron auth, each with a dedicated secret and constant-time comparison to prevent timing attacks
- Designed a multi-tenant data isolation architecture by enforcing tenant-scoped queries at the ORM layer across every API endpoint, verified via 6 automated integration tests confirming **zero cross-tenant data leakage**
- Evolved a production PostgreSQL schema through 9 zero-downtime Alembic migrations, including safe handling of Postgres enum-type constraints unsupported by migration autogeneration, applied in an isolated transaction

AccessForge | *FastAPI, LangGraph, PostgreSQL, Redis/RQ, Docker, Playwright, Axe-core*

- Developed an end-to-end accessibility remediation platform that scans React apps, generates and independently verifies WCAG fixes, and opens GitHub PRs, designed as a deterministic state machine using LangGraph
- Architected a crash-recoverable async pipeline using PostgreSQL-backed state and Redis workers, verified via a real worker crash captured during development, resuming task with **0 duplicate GitHub branches** or lost progress
- Validated LLM-generated accessibility fixes across 2 independent React/Vite codebases, including an independently built, real production app, using multimodal visual-regression comparison of rendered before/after screenshots
- Designed a confidence-gated localization check that deferred **68%** of detected violations to human review instead of guessing when no signal matched a source file, resolving **100%** of attempted accessible-name fixes on first pass

Work Experiences

Scientific Software Research Assistant — Jul. 2026 - Present
RCSB Protein Data Bank — *New Brunswick, NJ*

- Orchestrated a new MCP server tool exposing PDB search API to LLM agents enabling multi-service AND/OR queries in a single call, by composing the filter nodes from the 7 extended tools through a custom validation layer
- Rectified **2 silent data corruption bugs** in MCP server's result-grouping logic by tracing parameter flow against the tool's own docstrings, preventing the formation of invalid API parameters and user configurations

Summer Research Fellow (CSI SURF) — May. 2026 - Jul. 2026
Rutgers Artificial Intelligence and Data Science Collaboratory — *New Brunswick, NJ*

- Redesigned a dictionary-driven column-typing classifier in py-mmcif, an open-source Python package, cutting molecular structure file sizes by **27%** across an archive serving ∼**5M downloads** per day; shipped to production
- Optimized floating-point binary encoding in py-mmcif's compression pipeline, cutting compressed file sizes by up to **76%** across 37 varied PDB structures, working with a team of 4 researchers on the encoding chain design

Technical Skills

Languages: Python, Java, C, PHP, JavaScript (React, Next), SQL, HTML/CSS, Rust
Libraries/Frameworks: Tailwind CSS, Tauri, Flask, FastAPI, LangGraph, Playwright, Axe-core
Developer Tools: VS Code, IntelliJ, Git, Docker, WSL, Alembic, Neon, Cloudflare, MCP, Redis